

JONATHAN NADEAU

SUPPLY CHAIN • MANUFACTURING OPERATIONS • PROCESS AUTOMATION

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Portfolio Website: nadeaujonny.github.io

SUMMARY

Data analyst specializing in supply chain and manufacturing operations, with a track record of building automated procurement, inventory, and quality assurance tools that remove hundreds of hours of manual purchasing and engineering work per year. Sole developer of eight production tools at a defense and telecom electronics manufacturer, each owned from requirements through deployment. Mathematics graduate (UCSB), Google Data Analytics certified, ITAR eligible U.S. person.

SKILLS

Supply Chain & Operations: BOM analysis and costing, vendor sourcing and obsolescence risk, MRP simulation, ABC/XYZ segmentation, EOQ and safety stock, demand forecasting, inventory audit and reconciliation, ISO 9001 clause 9.1 reporting

Analysis & Reporting: Excel (Power Query, Power Pivot, DAX, Solver, FORECAST.ETS, dynamic arrays), Power BI, Tableau, Looker Studio

Databases & Programming: Oracle SQL, BigQuery, MySQL, SQLite • Python (pandas, openpyxl, scikit-learn, Streamlit, pytest, tkinter, PyInstaller), R

Systems & Integration: ETL pipelines, REST API and OAuth2 integration (DigiKey, Mouser, Bugzilla), ERP and legacy-system extraction, atomic and append-only file operations, Git

Quality & Communication: data validation and reconciliation, defect-injection and regression testing, requirements elicitation, technical writing, runbooks and user guides

AI-Assisted Development: Claude and Claude Code agentic workflows, spec-driven prototyping, work orders with gated verification, per-edit diff review, hand-authored test oracles; ships deterministic offline LLM-free binaries for regulated environments

SELECTED PORTFOLIO PROJECTS (15 deployed at nadeaujonny.github.io)

- Supply Chain Analytics (Oracle SQL, Power Query, Excel, Power BI):** Closed-loop pipeline over ~180K order records: normalized schema with stored procedures and a nightly MRP net-requirements simulation, ABC/XYZ classification, ETS demand forecasting with Solver-optimized safety stock and EOQ, forecast write-back into Oracle so recommendations take effect on the next cycle, and a six-page Power BI dashboard on a star schema with DAX and row-level security.
- Skechers (SKX) FP&A Model (Excel):** Fully articulating three-statement model, FY2020 to FY2024 actuals reconstructed from SEC 10-K filings plus a three-year forecast. Segment-level revenue engine, CHOOSE/MATCH Bear/Base/Bull scenario switch, two-variable sensitivity tables, and a KPI dashboard that verifies the statement tie rather than forcing it.
- Customer Churn Prediction (Python, scikit-learn, Streamlit):** End-to-end ML pipeline on 7,000+ records comparing five classifiers. Selected logistic regression at 0.83 AUC and 79% recall, rejecting a tuned model that traded 22 points of recall for a rounding-error AUC gain, with SHAP global and per-customer explanation. Deployed as an interactive app.

PROFESSIONAL EXPERIENCE

Data Analyst & Application Developer Intern | Escape Communications

July 2026 – Present | Torrance, CA

Sole developer on eight internal tools spanning procurement, inventory, quality, and engineering verification. Own requirements, architecture, testing, packaging, and deployment for each; report to the VP of Engineering.

- BOM Sourcing & Purchase Planning Tool:** Prices and risk-assesses entire bills of materials against live DigiKey and Mouser APIs, replacing the VP of Engineering's manual per-part lookup and spreadsheet cost build-up: an estimated 13 hours per BOM (300 to 400 parts at ~2 min each, plus ~1 hr of vendor data entry and calculation setup) now runs in about 3 minutes, reporting obsolescence risk, build cost, and restock timing. Diagnosed a suspected vendor data reliability problem as a packaging-variant defect, raising priced coverage from 65 to 195 of 266 rows; 703 passing tests.
- BOM Checker:** Audits PCB bills of materials for 16 classes of structural and parametric defect, decoding manufacturer part numbers offline to verify stated specs against the part actually called out. Surfaces upstream design errors that previously went unnoticed until late in the build cycle and removes an estimated 2 hours of row-by-row verification per BOM; validated on 8 production BOMs (2,290 rows, 869 part numbers); 427 tests including 26 defect-injection cases.
- Failure Analysis Search Tool:** Built REST ingestion, a Python extraction pipeline, and a Streamlit interface over 4,038 undocumented hardware failure records spanning 2007 to 2026, indexing 2,245 reference designators from 13,252 free-text engineer comments. Cut typical query time from roughly 6 seconds to 0.5, a 92% reduction, with TF-IDF similarity search and incremental watermark-tracked refresh.
- ERP Quality Report Generator:** Reverse-engineered six undocumented ISO 9001 clause 9.1 management-review reports from 20 archived workbooks and 16 database schemas, reproducing four consecutive years of results exactly (281/146/106/90 records). Automated generation saves the design engineer roughly 4 hours per reporting cycle; the audit surfaced 4 material errors, including a prior-year figure wrong by 3.4x carried forward two years.
- Common Parts Finder:** Identifies components shared across any number of BOMs, replacing side-by-side manual cross-checking of 100 to 390 row spreadsheets and saving the VP of Engineering an estimated 2 hours per comparison. Profiled 10 production BOMs first to surface five unstated requirements, including a matching rule that would have silently dropped 20 shared parts; requirements to deployable executable in one day.
- Inventory Audit Log Tool:** Consolidated physical inventory counts previously scattered across separate dated spreadsheets into one running audit log, one row per part and one column per audit date, eliminating the file-by-file reconciliation needed to trace how a part's stock moved. Append-only atomic writes guarantee a failed run leaves the record byte-identical; 69 tests.
- FPGA Waveform Capture Validation Tool:** Replaced by-eye waveform inspection with an automated validator that reconstructs loopback message streams from vendor capture archives and runs 13 named checks, letting the FPGA engineer assess modulation captures with materially greater accuracy by pinpointing the exact word and sample index of any mismatch. Identified two undocumented hardware handshake behaviors.
- Interface Documentation Generator:** Generates formatted Word interface documents from the same machine-generated JSON that drives a C/C++ code generator, removing the manual JSON parsing the embedded software engineer performed on every delivery and keeping documentation from drifting out of sync with the code; 139 tests against a hand-authored oracle predating the codebase.

Mathematics Instructor & Tutor | Mr. J's Academy • Wyzant • Blue Train College Prep

Sept 2022 – May 2025 | Torrance & Goleta, CA

Delivered 1,500+ hours of instruction to 17+ recurring clients, translating abstract quantitative material from algebra through calculus into terms non-technical audiences could act on: diagnosed each client's gap, built targeted materials, and adapted explanations in real time to the audience, the same skill applied to walking decision-makers through data and model output. Held a 4.8/5.0 rating and 100% client retention on Wyzant; promoted to high school lead mathematics specialist with a 50% compensation increase.

Lab Technician | Escape Communications

Summers 2017 – 2023, 2025 | Torrance, CA

Performed QA testing and calibration on power supply units: applied test loads, read multimeter outputs, and logged measurements into Excel QA trackers built with conditional formatting and logical formulas that automatically flagged out-of-tolerance readings. Assembled hardware units, tracked physical inventory across product lines, and functionally tested completed boards for compliance before release. This shop-floor exposure to BOMs, part numbering, and inventory practice is what later made the procurement and quality tooling above possible to specify correctly.

EDUCATION & CERTIFICATIONS

B.A. in Mathematics | University of California, Santa Barbara

Sept 2018 – March 2023

GPA 3.40 (Mathematics GPA 3.49) • 4x Dean's Honors • Minor: Secondary Mathematics & Science Education

Data Science Certificate | UCLA Extension • Python, R, SQL, Tableau, Spark, scikit-learn

June 2026 – Dec 2026 (in progress)

Google Data Analytics Professional Certificate | Coursera • 100% average, all nine courses

Oct 2025 – Jan 2026

Single Subject Mathematics Teaching Credential Coursework | CSU Long Beach • GPA 4.00

Aug 2024 – Oct 2025